UNIT I: Introduction

Topics Covered: AI problems, Intelligent agents (Agents and Environments, Rationality, Nature of environments, Structure of agents, Problem-solving agents, Problem formulation), Planning Application - Classical Planning problem. [1]

- **Artificial Intelligence (AI) Defined:** AI is the intelligence demonstrated by machines, a branch of computer science focused on creating intelligent machines. It aims to make computers perform tasks that humans currently do better, such as speech recognition, decision-making, and learning[2222222].
- **Intelligence:** The ability to acquire, understand, and apply knowledge to achieve goals[3].
- **Views of AI:** AI can be categorized into four views: thinking humanly, thinking rationally, acting humanly, and acting rationally[4].
    - **Acting Humanly (Turing Test):** Proposed by Turing (1950), it's an operational test where a computer passes if a human interrogator cannot distinguish its written responses from a human's[5].
- **AI Prehistory:** AI's roots are in philosophy (logic, reasoning, mind as a physical system), mathematics (algorithms, computation, probability), economics (utility, decision theory), neuroscience (physical substrate for mental activity), psychology (perception, motor control), computer engineering (building fast computers), control theory (designing systems that maximize objective functions over time), and linguistics (knowledge representation, grammar)[6].
- **Key Historical Milestones:**
    - **1943:** McCulloch & Pitts proposed the Boolean circuit model of the brain[7].
    - **1950:** Turing's "Computing Machinery and Intelligence" introduced the Turing Test[8].
    - **1956:** The term "Artificial Intelligence" was coined at the Dartmouth meeting[9].
- **AI Role in Mechanical Engineering:** AI impacts various aspects, including predictive monitoring, forecasting equipment breakdowns, product inspection, quality control, and assisting in Computer Integrated Manufacturing (CIM) and intelligent CAD systems[101010]. AI can reduce costs in semiconductor manufacturing and cooling systems (e.g., Google's Deepmind reduced cooling bills by 40% using ML algorithms)[11].
- **Agents and Environments:**
    - **Agent:** Anything that can perceive its environment through sensors and act upon that environment through effectors[12].
    - **Percept:** An agent's perceptual input at any given moment[13].
    - **Percept Sequence:** The complete history of everything the agent has ever perceived[14].
    - **Agent Function:** Maps any given percept sequence to an action[15].
    - **Agent Program:** Implements the agent function[16].
- **Rationality:** A rational agent acts to achieve the best outcome or, when uncertainty exists, the best expected outcome[17].
- **PEAS (Performance, Environment, Actuators, Sensors) Description:** A framework for specifying a task environment[18].
    - **Performance Measure:** Criteria for the agent's success[19].
    - **Environment:** The world in which the agent operates[20].
    - **Actuators:** Mechanisms for the agent to act[21].
    - **Sensors:** Mechanisms for the agent to perceive[22].

- **Environment Types:** Environments can be characterized by various properties:
  - **Fully Observable vs. Partially Observable:** An environment is fully observable if an agent's sensors give it access to the complete state of the environment[23].
  - **Single Agent vs. Multi-Agent:** Involves one or more agents[24].
  - **Deterministic vs. Stochastic:** Deterministic if the next state is completely determined by the current state and agent's action; otherwise, stochastic[25].
  - **Episodic vs. Sequential:** Episodic means each episode is independent; sequential means current action affects future decisions[26].
  - **Static vs. Dynamic:** Static if the environment doesn't change while the agent is deliberating; otherwise, dynamic[27].
  - **Discrete vs. Continuous:** Refers to the nature of states, time, and percepts/actions[28].
  - **Known vs. Unknown:** Known if the agent knows the outcomes of all actions[29].
- **Structure of Agents:**
  - **Simple Reflex Agents:** Select actions based on the current percept, ignoring percept history[30].
  - **Model-based Reflex Agents:** Maintain an internal state (model) of the world to handle partial observability[31].
  - **Goal-based Agents:** Use goal information to choose actions that lead to desired states[32].
  - **Utility-based Agents:** Choose actions that maximize their "utility" or happiness[33].
  - **Learning Agents:** Improve their performance over time[34].
- **Problem-Solving Agents:** These agents use search algorithms to find a sequence of actions that leads to a goal state[35353535].
- **Basic Search Strategies:**
  - **Problem Spaces:** Represent the problem in terms of states and actions[36].
  - **Uninformed Search (Blind Search):** Do not use any domain-specific knowledge beyond the problem definition[37373737].
    - **Breadth-First Search (BFS):** Explores all nodes at the present depth level before moving on to nodes at the next depth level. It is complete and optimal for unit step costs[383838383838383838].
    - **Depth-First Search (DFS):** Explores as far as possible along each branch before backtracking. It is not complete or optimal in general but can be memory-efficient[39393939393939].
    - **Depth-First with Iterative Deepening (IDDFS):** Combines DFS's space-efficiency with BFS's completeness and optimality. It performs a series of DFSs with increasing depth limits[40].
  - **Heuristic Search (Informed Search):** Uses problem-specific knowledge (heuristics) to guide the search[4141414141].
    - **Greedy Best-First Search:** Expands the node that appears to be closest to the goal (using a heuristic function h(n))[42424242]. It is not optimal or complete but can be very efficient[4343434343].
    - **A* Search:** Combines the cost to reach the current node g(n) with the estimated cost to reach the goal h(n). It is complete and optimal if the heuristic is admissible (never overestimates the cost to reach the goal)[44444444444444444444444444444444].

- **Admissible Heuristic:** A heuristic function h(n) is admissible if for every node n, h(n) is less than or equal to the true cost h*(n) to reach the goal from n[45].
- **Consistent Heuristic:** A heuristic h(n) is consistent if for every node n and every successor n' of n with step cost c(n, n'), h(n) <= c(n, n') + h(n')[46]. Consistency implies admissibility. [47]

UNIT II: Searching and Game Theory

Topics Covered: Searching for solutions, Searching with partial information (Heuristic search), Greedy best-first search, A* search, Constraint Satisfaction problem - Game Playing: Adversarial search, Games, Minimax algorithm, Optimal decisions in multiplayer games, Alpha-Beta pruning, Evaluation functions. Case studies: Tic-tac-toe game. [48]

- **Search Problem Components:**
  - **Search Space:** Set of possible solutions[49].
  - **Start State:** Where the agent begins[50].
  - **Goal Test:** Checks if the goal state is achieved[51].
  - **Search Tree:** A tree representation of the search problem[52].
  - **Actions:** Available actions for the agent[53].
  - **Transition Model:** Describes what each action does[54].
  - **Path Cost:** Numeric cost assigned to each path[55].
  - **Solution:** An action sequence from start to goal[56].
  - **Optimal Solution:** The solution with the lowest cost[57].
- **Properties of Search Algorithms:**
  - **Completeness:** Guarantees a solution if one exists[58].
  - **Optimality:** Guarantees the best solution (lowest path cost)[59].
  - **Time Complexity:** Time taken by the algorithm[60].
  - **Space Complexity:** Memory used by the algorithm[61].
- **Constraint Satisfaction Problem (CSP):**
  - A mathematical problem to find states or objects that satisfy a number of constraints[62].
  - Composed of a finite set of variables, each with a finite domain of values, and a set of constraints[63].
  - The task is to find an assignment of values to variables that satisfy all constraints[64].
  - **Backtracking Search:** A general algorithm for solving CSPs by incrementally building a solution and abandoning a partial solution when it violates a constraint[6565].
  - **Local Search for CSPs:** Starts with a complete but potentially invalid assignment and iteratively modifies it to satisfy constraints[6666].
- **Game Playing / Adversarial Search:**
  - **Games:** Strategic interactions where the outcome depends on the actions of multiple players[67].
  - **Minimax Algorithm:** An optimal strategy for two-player zero-sum games (where one player's gain is the other's loss), assuming both players play optimally[686868]. It aims to maximize the minimum possible gain[69].

        - **Minimax Value:** The best achievable utility against an optimal adversary[70].
        - It follows a Depth-First Search approach[71].
        - Propagates minimax values up the tree from terminal nodes[72].
    - **Optimal Decisions in Multiplayer Games:** The minimax principle can be extended to multiplayer games[73].
    - **Alpha-Beta Pruning:** An optimization technique for minimax search that eliminates branches of the search tree that cannot possibly influence the final decision[74747474].
        - **Alpha (α):** The best (highest) score that the maximizer can guarantee at a given node or above[75].
        - **Beta (β):** The best (lowest) score that the minimizer can guarantee at a given node or above[76].
        - Pruning occurs when α >= β[77].
        - **Move Ordering:** The effectiveness of alpha-beta pruning depends heavily on the order in which nodes are examined. Ideal ordering (best moves on the left) leads to more pruning[78].
    - **Evaluation Functions:** Heuristic functions used in game-playing algorithms to estimate the value of non-terminal states, especially when the search depth is limited[79].

UNIT III: Knowledge Representation and Reasoning

Topics Covered: Logical Agents, Knowledge-Based Agents, Wumpus world, Propositional logic, Resolution patterns in propositional Logic, First-order logic, Inference in first-order logic, Propositional vs. First-order inference, Unification and Lifting, Fo[80]rward chaining, Backward chaining, Resolution. [81]

- **Knowledge-Based Agents (KBAs):** Intelligent agents capable of maintaining an internal state of knowledge, reasoning over that knowledge, updating it based on observations, and taking actions efficiently[82828282].
    - **Components:** Knowledge-base (KB) and Inference System[838383].
    - **Knowledge-base:** A set of sentences representing what the agent knows about the world[84].
    - **Inference System:** Derives new sentences from old ones, applying logical rules to the KB to deduce new information[85].
    - **Operations:**
        - **TELL:** Informs the KB about new percepts from the environment[86].
        - **ASK:** Queries the KB to deduce appropriate actions[87].
        - **PERFORM:** Executes the action deduced by the agent[88].
- **Architecture of a KBA (Flowchart):**
    - Environment -> Input from Environment -> Inference Engine <-> Knowledge Base <- Learning (Updating KB) -> Inference Engine -> Output -> KBA [89]
- **Wumpus World:** A classic AI problem used to illustrate logical agents and knowledge representation. It's a grid-based game where an agent navigates a cave, avoiding pitfalls and a Wumpus, to find gold[90].
- **Propositional Logic (PL):** A simple form of logic where statements are propositions (declarative sentences that are either true or false). It uses logical connectives[91].
    - **Connectives:**
        - **AND (∧):** Conjunction[92].

        - **OR (∨):** Disjunction[93].
        - **NOT (¬):** Negation[94].
        - **Implication (→):** If-then[95].
        - **Biconditional (↔):** If and only if[96].
    - **Truth Table:** Defines the truth value of a compound proposition based on the truth values of its components[97].
    - **Logical Equivalence (⇔):** Two propositions are logically equivalent if they have the same truth values in all possible interpretations[98].
    - **Properties of Operators:** Commutativity, Associativity, Identity element, Distributive, De Morgan's Law, Double-negation elimination[99].
    - **Limitations of PL:** Limited expressive power; cannot represent relations like "ALL" or "SOME"[100].
- **First-Order Logic (FOL) / Predicate Logic:** An extension of propositional logic that allows for quantifiers (∀ - for all, ∃ - there exists) and predicates, enabling representation of relations and properties of objects[101].
    - **Components:** Constants, Predicates, Functions, Variables, Connectives, Quantifiers[102].
    - **Inference in FOL:** Rules for deriving new logical sentences from existing ones[103].
    - **Unification:** A process that finds substitutions for variables that make two logical expressions identical[104].
    - **Lifting:** Applying inference rules to first-order logic by using unification to find appropriate substitutions[105].
- **Inference Mechanisms:**
    - **Forward Chaining:** Starts with known facts and applies inference rules to derive new conclusions until the goal is reached[106]. It is data-driven[107].
    - **Backward Chaining:** Starts from the goal and works backward, looking for facts and rules that could support the goal[108]. It is goal-driven[109].
    - **Resolution:** A complete inference rule for propositional and first-order logic, used to determine if a conclusion can be logically derived from a set of premises[110]. It converts statements into Conjunctive Normal Form (CNF)[111].

UNIT IV: Uncertain Knowledge and Reasoning

Topics Covered: Bayes Rule, Concepts of Time and Uncertainty, Utility Functions, Value of Information, Value iteration, Policy iteration, Partially Observable MDP. [112]

- **Uncertainty:** Dealing with situations where complete knowledge is unavailable[113].
    - **Sources of Uncertainty:** Laziness (too much effort for complete knowledge), Theoretical ignorance (no complete theory), Practical ignorance (lack of data)[114].
- **Probability:** The most common approach to handling uncertainty, quantifying an agent's beliefs[115].
    - **Prior Probability:** Degree of belief before any evidence[116].
    - **Conditional Probability (Posterior Probability):** Probability of an event given that another event has occurred[117].
- **Bayes' Rule (Bayes' Theorem):** A fundamental theorem in probability that describes how to update the probability of a hypothesis based on new evidence[118118].

- Formula: P(A|B) = [P(B|A) * P(A)] / P(B) [119]

- **Utility Functions:** Quantify the desirability of different outcomes, often used in decision-making under uncertainty[120].
  - **Principle of Maximum Expected Utility (MEU):** A rational agent chooses the action that maximizes its expected utility[121].
  - **Lottery:** A representation of uncertain outcomes, where outcomes occur with certain probabilities[122].
- **Value of Information:** The expected increase in utility from gaining new information before making a decision[123].
- **Markov Decision Processes (MDPs):** A mathematical framework for modeling decision-making in situations where outcomes are partly random and partly under the control of a decision-maker[124].
  - **Components:** Set of states, set of actions, transition model (probability of reaching a state given current state and action), reward function, and discount factor[125125].
- **Value Iteration:** An algorithm to find the optimal policy for an MDP by iteratively computing the optimal value function for each state[126].
- **Policy Iteration:** An algorithm that alternates between policy evaluation (calculating the value of the current policy) and policy improvement (finding a better policy based on the evaluated values)[127].
- **Partially Observable Markov Decision Process (POMDP):** A framework for decision-making under uncertainty where the agent does not have complete or perfect information about the current state of the environment[128].
  - The agent maintains a "belief" (probability distribution) over possible states, updated using Bayes' rule with new observations[129].

UNIT V: Basics of Deep Learning

Topics Covered: Deep learning architectures, Convolutional Neural Networks (CNNs): Neurons in Human Vision, The Shortcomings of Feature Selection, Full Description of the Convolutional Layer, Max Pooling, Full Architectural Description of Convolution Networks, Closing the Loop on MNIST with Convolutional Networks, Building a Convolutional Network for CIFAR-10, Visualizing Learning in Convolutional Networks, Leveraging Convolutional Filters to Replicate Artistic Styles, Learning Convolutional Filters for Other Problem [130]Domains, Training algorithms. [131]

- **Deep Learning:** A subfield of Machine Learning that uses artificial neural networks with multiple layers (deep neural networks) to learn representations of data with multiple levels of abstraction[132132132132132132132].
  - **Artificial Neural Networks (ANNs):** Computational models inspired by the structure and function of biological neural networks, consisting of interconnected nodes (neurons) organized in layers[133].
  - **Simple Deep Learning Network Architecture (Flowchart):**
    - Input Layer -> Hidden Layer(s) -> Output Layer [134]
- **Convolutional Neural Networks (CNNs):** A class of deep neural networks specifically designed for processing structured grid-like data, such as images. They are highly effective for tasks like image classification, object detection, and image generation[135].
  - **Neurons in Human Vision:** CNNs are inspired by the organization of the visual cortex in animals, where individual neurons respond to specific regions of the visual field[136].
  - **Shortcomings of Feature Selection:** Traditional machine learning often relies on manual feature engineering, which is time-consuming and may not capture complex patterns. CNNs overcome this by automatically learning relevant features from data[137].

- **Convolutional Layer:** The core building block of a CNN. It applies a set of learnable filters (kernels) to the input data to produce feature maps[138].
    - **Filters/Kernels:** Small matrices that slide across the input, performing element-wise multiplications and summing the results to detect patterns like edges, textures, or more complex shapes[139].
- **Max Pooling:** A down-sampling operation that reduces the spatial dimensions of the feature maps, helping to make the network more robust to small translations and reducing computational cost[140]. It typically selects the maximum value within a filter's receptive field[141].
- **Full Architectural Description of Convolution Networks:** Typically consist of alternating convolutional and pooling layers, followed by fully connected layers for classification or regression[142].
- **VGGNet:** A well-known deep CNN architecture, notable for its depth (e.g., VGG16, VGG19 with 16 and 19 weight layers respectively) and uniform use of 3x3 convolution filters[143].
- **Applications of CNNs:**
    - **Image Classification:** Identifying the main object or category in an image[144].
    - **Object Recognition:** Detecting and localizing multiple objects within an image[145].
    - **Facial Recognition:** Identifying individuals from images[146].
    - **Image Synthesis:** Generating new images, such as texture synthesis[147].
    - **Artistic Style Transfer:** Leveraging convolutional filters to apply the artistic style of one image to the content of another[148].
    - **Image Colorization:** Training CNNs to colorize grayscale images[149].
    - **Filter Removal:** Reversing artistic filters applied to images[150].
- **Training Algorithms:** CNNs are typically trained using backpropagation and optimization algorithms like stochastic gradient descent (SGD) to adjust the network's weights and biases based on the error between predicted and actual outputs[151].

UNIT VI: Deep Reinforcement Learning

Topics Covered: Deep Reinforcement Learning Masters Atari Games, Reinforcement Learning, Markov Decision Processes (MDP), Explore Versus Exploit, Pole-Cart with Policy Gradients, Q-Learning and Deep Q-Networks, Improving and Moving Beyond DQN. [152]

- **Reinforcement Learning (RL):** A branch of machine learning where an agent learns to make decisions by interacting with an environment, receiving feedback in the form of rewards or penalties, and aiming to maximize cumulative reward over time[153153153153].
    - **Key Elements:** Agent, Environment, States, Actions, Rewards, Policy, Value Function[154].
    - **Policy:** The agent's strategy for choosing actions in different states[155].
    - **Value Function:** Estimates the long-term desirability of states or state-action pairs[156].
- **Markov Decision Processes (MDPs):** (As described in Unit IV) The formal framework for reinforcement learning problems[157157157].
- **Explore Versus Exploit:** A fundamental dilemma in RL where the agent must balance exploring new actions to discover better rewards with exploiting known good actions to maximize current rewards[158158158].
- **Deep Reinforcement Learning (DRL):** Combines deep neural networks with reinforcement learning, allowing agents to learn directly from raw sensory input (like pixels in games)[159].

    - **Deep Reinforcement Learning Masters Atari Games:** A major breakthrough in 2014 by DeepMind, where a Deep Q-Network (DQN) learned to play 49 different Atari games with superhuman skill using the same architecture[160160].

- **Pole-Cart with Policy Gradients:** An example of using policy gradient methods in DRL. Policy gradient algorithms directly learn a policy that maps states to actions, often used in continuous action spaces[161].
- **Q-Learning:** A model-free reinforcement learning algorithm that learns an action-value function (Q-function), which gives the expected utility of taking a given action in a given state[162162162].
    - **Q-Value (Q(s,a)):** Represents the expected future reward for taking action a in state s and then following the optimal policy thereafter[163].
    - **Bellman Equation:** Defines the relationship between Q-values, stating that the maximum future reward for taking an action is the current reward plus the next step's max future reward[164].
    - **Value Iteration:** The process of updating Q-values based on future Q-values[165].
- **Deep Q-Networks (DQN):** A type of Q-learning that uses a deep neural network to approximate the Q-function, enabling it to handle high-dimensional state spaces (e.g., raw image pixels)[166166166].
- **Improving and Moving Beyond DQN:** Further advancements in DRL beyond basic DQN, such as Double DQN, Dueling DQN, Prioritized Experience Replay, and others, aim to improve stability, sample efficiency, and performance[167].

Please let me know if you'd like a deeper dive into any specific topic or if you have further questions!